Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Lesson 1: Model Selection Techniques

- Introduction to Model Selection
  Reading • 5 min
- Bayesian Model Selection
  Reading • 30 min
- Bayesian Model Selection Readings
  Reading • 1h

Lesson 2: Cross Validation

- Introduction to Cross-Validation
  Reading • 25 min
- K-Fold Cross-Validation
  Reading • 2 min
- Leave-One-Out Cross-Validation (LOOCV)
  Reading • 10 min

Lesson 3: Introduction to Optimization Techniques

- Introduction to Optimization Techniques
  Reading • 13 min
- **Optimization Using Gradient Descent**
  Reading • 40 min
- Gradient Descent with Momentum
  Reading • 45 min
- Stochastic Gradient Descent (SGD)
  Reading • 40 min
- Constrained Optimization
  Reading • 16 min
- Lagrange Multipliers
  Reading • 8 min
- Convex Optimization
  Reading • 15 min
- Linear Programming
  Reading • 10 min
- Quadratic Programming
  Reading • 10 min
- Check Your Knowledge: Optimization Techniques
  Graded Assignment • 15 min

Optimization Using Gradient Descent

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Gradient Descent is a first-order optimization algorithm widely used in machine learning to minimize a differentiable objective function

$$minf(x)$$

$f : \mathbb{R}^d \to \mathbb{R}$, where d represents the number of dimensions or parameters. The goal is to find a local minimum of the function, which corresponds to the optimal parameters of the model.

Objective Function and Differentiability

We assume that $f(x)$ is differentiable, meaning that it has a gradient at every point. However, in many cases, solving this optimization problem analytically (finding a closed-form solution) is not feasible due to the complexity of the function.

Gradient Descent

Gradient Descent is a technique used to iteratively approach a local minimum of the function $f(x)$. The fundamental idea is to start from an initial guess of the parameters x_0 and then iteratively update these parameters by moving in the direction opposite to the gradient of the function at the current point.

The gradient of the function $\nabla f(x)$ represents the direction of the steepest ascent, so moving in the negative direction of this gradient corresponds to moving in the direction of the steepest descent.

Mathematically, this is represented as:

$$x_{i+1} = x_i - \gamma_i(\nabla f\,(x_i))^T$$

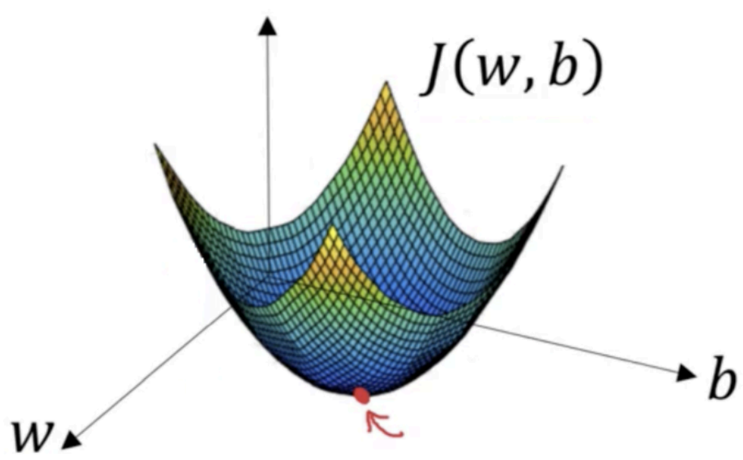
Where:

- x_i is the current parameter vector at iteration i.
- γ_i is the step-size or learning rate, controlling the size of the steps.
- $\nabla f(x_i)$ is the gradient of $f(x)$ at x_i.
- $\nabla f(x_i)^T$ denotes the transpose of the gradient vector to ensure the dimensions are compatible for subtraction.

Gradient and Contour Lines

To visualize this, imagine the function $f(x)$ as a surface with hills and valleys. The gradient at a point x_0 points in the direction of the steepest ascent, orthogonal to the contour lines of the function; these contour lines represent the set of points where the function takes the same value, $f(x) = c$. By moving in the direction opposite to the gradient, we effectively roll a ball down the surface, always heading towards the steepest decline.

Resource: Gradient Descent - BuiltIn (30 minutes)



Iterative Process

The iterative nature of gradient descent means that starting from an initial guess x_0, the algorithm repeatedly updates the parameters according to the above formula.

For a suitably chosen step size γ_i, the sequence of function values $f(x_0)$, $f(x_1)$, $f(x_2)$, ….. will decrease, ideally converging to a local minimum $f(x^*)$.

Choosing the Step Size γ

The step size γ is crucial for the convergence of gradient descent. If γ is too large, the algorithm might overshoot and fail to converge. If γ is too small, the algorithm will converge slowly, making the process inefficient.

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